

School of Computer Science and Artificial Intelligence

Lab Assignment # 4

Name of Student : CHATHARABOINA SAGAR
Enrollment No. : 2303A51522
Batch No. : 22

Task-1:

Write a Python function to determine whether a given number is prime

Expected Output-1:

A basic Python function to check if a number is prime, demonstrating correct logical conditions without relying on examples or additional context

Prompt:-

write a python code function to determine whether the number is prime

Code:-

```
lab4.py > ...
1  #write a python code function to derermine wether the number is prime
2  def is_prime(n):
3      if n <= 1:
4          return False
5      for i in range(2, int(n**0.5) + 1):
6          if n % i == 0:
7              return False
8      return True
9  # Example usage
10 number = int(input())
11 if is_prime(number):
12     print(f"{number} is a prime number.")
13 else:
14     print(f"{number} is not a prime number.")
```

Output:-

```
PS C:\sagar\pack1\OneDrive\3-2\AI Assistant Coding> & C:/Users/dell/AppData/Local/Programs/Python/Python312/python.exe "c:/sagar/pack1/OneDrive/3-2/AI Assistant Coding/lab4.py"
2
2 is a prime number.
```

Observation

- The function is correct but basic.
- Logic is simple without optimization.

Task Description – 2**Input:**

[1, 2, 3, 4], Output: 10 to help AI generate a function that calculates the sum of elements in a list.

Expected Output-2

- A correct conversion function guided by the single example.

Prompt:-

write a python code a function calculates the sum of elements in a list

Example usage input format for the user

Code:-

```
#--Task 2--
#write a python code a function claculates the sum of elements in a list
def sum_of_list(lst):
    total = 0
    for num in lst:
        total += num
    return total

# Example usage input format for the user
input_list = list(map(int, input().split()))
result = sum_of_list(input_list)
print(f"The sum of the list elements is: {result}")
```

Output:

```
1 2 3 4
The sum of the list elements is: 10
PS C:\sagar\pack1\OneDrive\3-2\AI Assistant Coding> |
```

Observation

- The example helps AI understand expected behaviour.
- Output accuracy improves compared to zero-shot.

Task Description – 3

- Few-shot: Give 2–3 examples to create a function that extracts digits from an alphanumeric string.

Expected Output-3

- Accurate function that returns only the digits from alphanumeric string.

Prompt:-

write a python code a function to the extract digits from an alphanumeric string.

example for user in input format

Code:

```
#---Task-3
#write a python code a function to the extract digits from an alphanumeric string.
#example for user in input format
def extract_digits(s):
    digits = ''.join([char for char in s if char.isdigit()])
    return digits
input_string = input()
result = extract_digits(input_string)
print(f"Extracted digits: {result}")
```

Output:

```
a3a4n5nf5
Extracted digits: 3455
PS C:\sagar\pack1\OneDrive\3-2\AI Assistant Coding>
```

Observation

- Code is accurate and reliable.
- Multiple examples reduce ambiguity.

Task Description – 4

- Compare zero-shot vs few-shot prompting for generating a function that counts the number of vowels in a string.

Expected Output-3

- Output comparison + student explanation on how examples helped the model.

Prompt:

Write a python code a function to count vowels in a given string.
by the input format for the user

Code:

```
--Task-4--
#Write a python code a function to count vowels in a given string.
#by the input format for the user
def count_vowels(s):
    vowels = 'aeiouAEIOU'
    count = sum(1 for char in s if char in vowels)
    return count
input_string = input()
vowel_count = count_vowels(input_string)
print(f"Number of vowels: {vowel_count}")
```

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Output:

```
My Name is Sagar  
Number of vowels: 5
```

Task Description – 5

Use few-shot prompting with 3 sample inputs to generate a function that determines the minimum of three numbers without using the built-in min() function.

Expected Output-5

- A function that handles all cases with correct logic based on example patterns.

Prompt:

Write a python code Minimum of three numbers by the in user input format

code:

```
#write a python code Minimum of three numbers  
#by the in user input format  
def min_of_three(a, b, c):  
    return min(a, b, c)  
a, b, c = map(int, input().split())  
minimum = min_of_three(a, b, c)  
print(f"The minimum of the three numbers is: {minimum}")
```

Output:

```
5 6 7  
The minimum of the three numbers is: 5
```